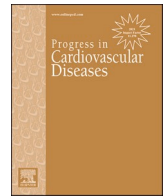




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Review Article

Digital care in your pocket: The role of technology in exercise, fitness, prevention, and rehabilitation

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ABSTRACT

Purpose: The digital transformation of healthcare is creating new opportunities to enhance access, engagement, and outcomes in exercise, fitness, cardiovascular disease (CVD) prevention, and rehabilitation. This review synthesizes current evidence on digital health technologies across the cardiovascular care continuum, emphasizing their role in promoting physical activity in both primary and secondary prevention.

Methods: A narrative review was conducted examining mobile health applications, wearable devices, artificial intelligence, remote monitoring platforms, and emerging technologies including virtual reality, 5G connectivity, and large language models. Applications were evaluated in the context of CVD prevention, rehabilitation, and behavior change.

Results: Digital tools can personalize exercise prescriptions, monitor physiologic metrics, and support remote supervision in clinical and community settings. Wearables and mobile platforms demonstrate utility in improving adherence, fitness, and CVD risk. AI-driven systems enable adaptive programming, patient monitoring, and predictive modeling, while virtual and augmented reality offer immersive options for home-based rehabilitation. Persistent barriers include digital literacy, inequitable access, and data privacy concerns.

Conclusion: Digital health technologies are reshaping exercise-based cardiovascular care. When coupled with behavior change strategies and ethical implementation, they can extend the reach and impact of physical activity interventions. Further research is needed to guide clinical integration, long-term evaluation, and equitable adoption.

Introduction

The health benefits of physical activity (PA) are well established across cardiovascular and preventive medicine, with consistent associations observed across more than 25 chronic disease states.¹ Regular physical activity is associated with substantial reductions in premature mortality and lower incidence of cardiovascular disease (CVD), type 2 diabetes, stroke, and hypertension, with effect sizes that often meet or exceed those of pharmacologic prevention strategies.^{2–6} Despite this strong evidence base, population-level physical activity remains persistently low. Globally, more than 70% of adults do not meet recommended physical activity levels, a pattern that has changed little over

recent decades.¹ This ongoing gap between evidence and real-world uptake represents a major challenge for cardiovascular prevention and rehabilitation.

Conventional approaches to physical activity promotion—centered on education, brief counseling, and episodic follow-up—have demonstrated limited effectiveness in achieving sustained behavior change, particularly in higher-risk populations. In parallel, CVD remains the leading cause of morbidity and mortality worldwide, reinforcing the need for scalable, durable, and evidence-based preventive strategies that extend beyond clinic-based care models.^{7,8} In response, professional organizations including the World Health Organization and the American Heart Association have increasingly emphasized the role of

Abbreviations: AI, Artificial Intelligence; API, Application Programming Interface; AR, Augmented Reality; CVD, Cardiovascular disease; GDPR, General Data Protection Regulation; HIPAA, Health Insurance Portability and Accountability Act; LLM, Large Language Model; VR, Virtual Reality.

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technology-enabled interventions to support long-term engagement, personalization, and continuity of care.^{9,10}

Recent advances in digital health have created new opportunities to operationalize these recommendations. Widespread adoption of wearable sensors, mobile health applications, and remote monitoring platforms now enables continuous capture of physiologic and behavioral data in real-world settings.^{11–14} These capabilities are increasingly augmented by artificial intelligence (AI)-driven analytics, which support individualized feedback, adaptive exercise prescription, and clinical decision support. At the systems level, improvements in data infrastructure, interoperability, and high-speed connectivity—including fifth generation (5G) wireless networks—facilitate scalable, responsive, and integrated delivery of digital cardiovascular care. Collectively, these developments have enabled a shift toward more continuous, personalized models of prevention and rehabilitation, effectively placing elements of cardiovascular care into patients' everyday environments.

This narrative review synthesizes current evidence on digital health technologies used to support physical activity across the cardiovascular care continuum, with a focus on both primary and secondary prevention. We examine enabling technologies, emerging applications, and implementation considerations relevant to exercise promotion, fitness, and cardiac rehabilitation. Fig. 1 provides a conceptual overview of the digital ecosystem underlying this paradigm, illustrating how data captured from wearables and connected devices are translated into personalized actions through software platforms, analytics, and clinical workflows.

The review is structured around key technology domains and their clinical applications. Table 1 outlines the core analytic technologies underlying digital exercise systems and is referenced throughout the following sections when discussing AI, machine learning, and computer vision applications. Table 2 summarizes software-based applications, while Table 3 details hardware components that enable remote monitoring and exercise delivery. Across these sections, we integrate evidence from clinical studies, systematic reviews, and real-world implementations, with attention to scalability, equity, and clinical integration.

Through this synthesis, we aim to clarify the evolving role of digital technologies in addressing long-standing barriers to physical activity and to inform their application in contemporary cardiovascular prevention and rehabilitation. This review is organized around three interrelated themes. First, we examine how digital tools, including mobile health platforms, wearable sensors, immersive technologies, and

Table 1

Core technologies used in digital exercise, fitness, and rehabilitation.

Name	Brief Definition	Example in Exercise, Fitness, and Rehabilitation
AI	A field of computer science that simulates human intelligence to perform tasks such as reasoning, learning, and decision-making.	AI-driven apps that provide adaptive exercise recommendations based on real-time user feedback.
Machine Learning	A subset of AI that uses algorithms to identify patterns in data and improve performance over time without explicit programming.	Algorithms that analyze user performance data to personalize workout routines.
Deep Learning	A subset of machine learning that uses multi-layered neural networks to model complex, high-dimensional patterns in large datasets.	Wearables that track physical activity and detect deviations using deep learning models.
Convolutional Neural Networks	A class of deep learning models particularly suited for analyzing visual data, such as images or video.	Video-based systems that assess posture and provide form corrections during remote rehabilitation.
Natural Language Processing	The ability of computers to interpret, understand, and generate human language in written or spoken form.	Virtual assistants that provide voice-guided exercise instruction and answer health-related questions.
Computer Vision	A field of AI that enables machines to interpret and act on visual inputs from the world, such as images or video streams.	Motion tracking applications that provide real-time feedback on movement and detect improper form during exercises.
Large Language Models (LLMs)	A subset of natural language processing that uses deep learning to generate human-like language by training on vast corpora of text. LLMs can understand, generate, and summarize natural language.	LLMs like ChatGPT support exercise coaching, answer health-related questions, generate personalized content, and improve health literacy.

Abbreviations: AI, artificial intelligence.

artificial intelligence, are reshaping exercise-based cardiovascular prevention and rehabilitation. Second, we describe the technological infrastructure that enables these tools, including data connectivity, analytics, and interoperability. Third, we address implementation considerations, including equity, reimbursement, workflow integration, and long-term sustainability. Together, these domains define both the present state and future trajectory of digital cardiovascular care.

Signal strength and bandwidth: The role of fifth-generation (5G) technology in healthcare

High-speed wireless connectivity functions as enabling infrastructure for digital cardiovascular care. Advances in fifth-generation (5G) networks improve bandwidth, reliability, and synchronization across connected devices, supporting scalable delivery of exercise, monitoring, and rehabilitation interventions beyond traditional clinical settings.^{15–19} Rather than fundamentally altering clinical decision-making workflows, enhanced connectivity primarily improves data completeness, interoperability, and user experience across digital platforms.^{13,14}

In digital cardiovascular applications, it is important to distinguish between bandwidth, latency, and synchronization. Bandwidth determines the volume of data that can be transmitted efficiently; latency reflects the delay between data transmission and receipt; and synchronization ensures coordinated communication among devices, applications, and cloud-based systems.^{17,18} These characteristics influence how

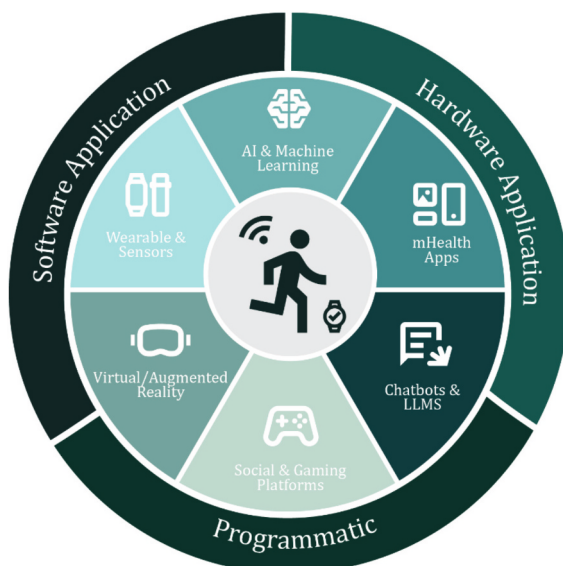


Fig. 1. Digital ecosystem of cardiovascular health and fitness.

Table 2
Software applications used in digital exercise, fitness, and rehabilitation.

Name	Brief Definition	Example in Exercise, Fitness, and Rehabilitation
Software as a Service	Cloud-based software delivery model that allows users to access applications over the internet on a subscription basis, without installing or maintaining local servers.	Platforms like Apple Fitness+ and Mia Health offer personalized workouts, progress tracking, and integration with wearables through cloud delivery.
Mobile Health Apps	Smartphone-based applications that deliver health-related services, including fitness tracking, symptom monitoring, and educational content.	Cardiac rehab apps deliver guided exercise videos, track steps, and offer heart rate monitoring in real time.
Virtual Reality (VR)	A fully immersive, computer-generated simulation that enables users to interact with a three-dimensional environment.	VR-based rehab tools simulate physical therapy routines or cycling environments to improve adherence in cardiac rehabilitation.
Augmented Reality (AR)	Technology that overlays virtual elements on the real world, enhancing the physical environment with interactive data or visual guides.	AR apps display animated trainers in the user's environment to demonstrate exercises and provide visual corrections in real time.
Chatbots	AI-driven programs designed to simulate human conversation via text or voice, offering support and automated responses.	Cardiac rehab chatbots remind patients to complete their workouts, track symptoms, and offer encouragement.
Application Programming Interface	A set of tools and protocols that enable different software systems to communicate and exchange data seamlessly.	Google Fit's and Mia Health's application programming interface allows third-party apps to integrate step counts and heart rate data for personalized fitness tracking.
Social Networks	Digital platforms that allow users to share information, communicate, and form online communities.	Private rehab communities offer peer support and motivation for patients completing remote exercise programs.
Gaming	Interactive software that incorporates game mechanics such as scoring, rewards, and challenges to promote engagement and behavior change.	Exergames like Ring Fit Adventure and Beat Saber combine fun with physical activity, improving cardiovascular endurance.

Abbreviations: VR, virtual reality; AR, augmented reality.

digital tools function in practice. For most prevention and rehabilitation applications—including blood pressure monitoring, activity tracking, and symptom reporting—periodic or asynchronous data transmission is sufficient and aligns with established remote patient monitoring models used in implantable cardiac devices and other clinical-grade systems.^{20,22,40,79}

Low-latency transmission becomes more critical in applications requiring synchronous interaction, particularly live video-based exercise supervision or virtual and augmented reality-supported rehabilitation. These modalities depend on stable, high-bandwidth connections to maintain audiovisual fidelity, motion tracking accuracy, and real-time feedback during supervised sessions.^{17,45–47} More broadly, improved connectivity supports integration across wearable sensors, mobile health platforms, electronic health records, and analytic systems, enabling scalable and coordinated delivery of digital cardiovascular care.^{13,14,78}

The rise of digital technologies in physical activity interventions

For the purposes of this narrative review, digital health refers to the integration of software applications, connected hardware, data

Table 3
Hardware applications used in digital exercise, fitness, and rehabilitation.

Name	Brief Definition	Example
Wearables	Body-worn devices equipped with sensors to collect real-time physiological and activity data.	Smartwatches track heart rate, electrocardiography (ECG), maximal oxygen uptake (VO ₂ max), and steps to support personalized exercise plans and remote rehabilitation.
Connected Devices	Standalone medical or wellness tools (e.g., blood pressure cuffs, glucometers) that sync with apps or systems to transmit health data.	Bluetooth-enabled blood pressure cuffs send readings to a mobile app, helping clinicians monitor hypertensive patients during exercise routines.
Smartphones	Mobile computing devices that run health applications and interface with sensors and digital platforms for care delivery.	Smartphones host cardiac rehab apps, display exercise guidance, and sync with wearables for real-time feedback and tracking.
Webcams	Camera hardware that enables visual interaction and remote supervision via telehealth platforms.	Webcams are used for virtual group fitness classes and tele-rehab, allowing providers to assess patient form and engagement.
Earbuds and Headsets	Audio devices used to deliver guided exercise prompts, coaching, and feedback, some embedded with biometric sensors.	Heart rate-monitoring earbuds provide verbal cues to help patients maintain safe intensity zones during cardiovascular workouts.
Peripherals (Mice & Keyboards)	Input devices used to navigate digital interfaces, especially on desktops or tablets.	Older adults use simplified keyboards and mice to access educational modules and virtual rehab platforms at home.
Transmission (Bluetooth & Cellular)	Wireless technologies that enable real-time data transfer between devices and cloud-based systems.	Wearables transmit exercise data via Bluetooth to smartphones, which upload it over cellular networks to providers' remote dashboards.
Infrastructure (Databases & Telecommunications)	The backend systems supporting data storage, processing, and communication across platforms.	Cloud-based databases store rehab progress and biometric trends, while telecom infrastructure ensures timely delivery of remote services.

Abbreviations: ECG, electrocardiography; VO₂ max, maximal oxygen uptake.

infrastructure, and analytic systems that enable the delivery, monitoring, and personalization of exercise-based cardiovascular prevention and rehabilitation beyond traditional in-person clinical encounters. This definition encompasses mobile health platforms, wearable sensors, immersive technologies, artificial intelligence-driven analytics, and the connectivity systems that allow these components to function as an integrated ecosystem.

Digital technologies are increasingly embedded in physical activity (PA) promotion, exercise training, and rehabilitation, reflecting a shift toward remote, personalized, and scalable care models. Adoption of consumer-facing digital health tools has grown substantially over the past decade. Wearable device use has increased by more than 20% annually in recent years, and mobile health and fitness applications represent one of the fastest-growing segments of the digital health market.^{24,25} These trends indicate a meaningful change in how individuals engage with PA outside traditional clinical or fitness settings.

The sections that follow illustrate how different categories of digital tools, wearables, mobile applications, online platforms, immersive technologies, and personalized analytic systems, contribute to a common objective: extending the reach, precision, and sustainability of exercise-based cardiovascular care. Although presented separately for

clarity, these tools typically function in combination rather than isolation.

Wearables and mobile applications

Wearable devices, including fitness trackers and smartwatches, are widely used to capture objective measures such as step count, heart rate, sleep, and energy expenditure. Systematic reviews demonstrate that commercially available wearables provide acceptable validity and reliability for step count and heart rate measurement, supporting their use in PA monitoring and behavior change interventions.²³ When integrated with mobile applications, these tools enable goal setting, trend visualization, and behavioral feedback—features associated with improved PA levels and adherence.²⁵

Hardware components outlined in Table 3, particularly wearable sensors, enable the physiologic data streams required for personalized exercise guidance. Fitness and training applications deliver structured exercise programs, guided workouts, and progress tracking, while complementary applications address nutrition, recovery, and stress management. Evidence from longitudinal and theory-based studies suggests that sustained app engagement is associated with higher activity levels and improved adherence, particularly when behavior change techniques are embedded into app design.^{26,27} Together, these technologies illustrate how objective physiologic monitoring and behavioral feedback form the foundation of data-driven exercise personalization.

Online platforms, social features, and gamification

Online fitness platforms extend instructor-led exercise into the home through live and on-demand programming, while social features such as challenges, leaderboards, and virtual communities enhance accountability and motivation. Gamification strategies—including goal-based rewards and competition—have been associated with higher engagement, enjoyment, and persistence in PA programs.^{25,28} Exergaming and virtual fitness applications further integrate physical activity with interactive experiences, supporting adherence in both recreational and rehabilitation contexts.²⁸

Immersive technologies and personalized exercise

Virtual and augmented reality (AR)-based exercise systems offer immersive environments that enhance engagement, particularly in supervised or semi-supervised settings. These applications rely on stable, high-bandwidth connectivity to support audiovisual synchronization and motion tracking during interactive exercise sessions. Advances in artificial intelligence further enable personalized exercise prescription by integrating user goals, preferences, and physiologic data to adapt training intensity and progression over time.²⁹ Immersive platforms represent an evolution from passive monitoring toward interactive, adaptive rehabilitation environments.

Adoption drivers and equity considerations

The uptake of digital PA technologies is driven by convenience, flexibility, and personalization, which reduce common barriers related to time, location, and access.²⁵ However, adoption remains uneven. Older adults, individuals with lower socioeconomic status, and those living in rural or underserved communities demonstrate lower utilization, reflecting disparities in access and digital literacy.²¹ Emerging evidence suggests that simplified interfaces, culturally tailored content, and hybrid models combining digital tools with human support may help mitigate these gaps, though further evaluation is needed.

Despite their potential, digital PA interventions raise ongoing concerns related to data privacy, security, and long-term engagement. Addressing these challenges will be critical to ensure that digital

technologies enhance physical activity promotion without exacerbating existing health disparities. Ultimately, the impact of digital health tools depends not only on technical capability but also on equitable access, usability, and integration into real-world care settings.

Collectively, these tools illustrate how software and hardware components translate analytic capability into patient-facing exercise interventions, reinforcing the central role of digital systems in extending cardiovascular prevention beyond clinic walls.

Health technology at the intersection of healthcare and computer science

AI is emerging as a transformative force in healthcare, reshaping approaches to prevention, diagnosis, rehabilitation, and chronic disease management—particularly in cardiovascular health. Although the concept of AI dates back several decades, recent advances in machine learning, deep learning, and cloud computing, coupled with access to large and diverse datasets, have accelerated its implementation in both clinical and consumer-facing environments.^{30,31}

AI refers to the simulation of human intelligence processes by computer systems, including learning, reasoning, problem-solving, and natural language understanding.^{30,32} It draws on multidisciplinary fields such as computer science, psychology, linguistics, and philosophy to design systems capable of performing complex tasks traditionally requiring human cognition. Among the AI modalities most relevant to health and fitness are machine learning, deep learning, natural language processing, neural networks, computer vision, voice recognition, and virtual coaching platforms.³³ These techniques support applications ranging from interpretation of electrocardiograms and medical imaging to conversational agents that promote patient engagement.^{20,27}

As summarized in Table 1, AI-driven analytic systems form the computational backbone of many digital exercise platforms. In exercise science and cardiac rehabilitation, AI offers significant advantages. Algorithms can personalize exercise recommendations by integrating real-time performance data, health history, and comorbidities to adjust exercise intensity, frequency, and modality.³⁴ Predictive analytics further enable anticipation of adverse events—such as arrhythmias or blood pressure instability—supporting earlier intervention by clinicians or digital platforms.³⁵

Clinical applications are expanding rapidly. In cardiac rehabilitation, natural language processing-powered virtual coaches support patients during remote recovery by adapting regimens based on physiologic responses and patient-reported outcomes. AI models are also used to guide safe exercise initiation and progression through cardiovascular risk stratification using integrated laboratory, imaging, and historical data.³⁶

The trajectory of AI in cardiovascular medicine increasingly emphasizes explainable AI approaches that enhance transparency and interpretability for clinicians and patients.³⁶ Concurrent advances in wearable biosensors and multimodal physiologic monitoring enable continuous data streams that support dynamic feedback loops for more responsive, personalized care. Ultimately, the most effective systems will combine AI's analytical capabilities with human clinical judgment, enabling hybrid care models that optimize safety, engagement, and outcomes.³²

Computer vision in cardiovascular fitness and rehabilitation

Computer vision, a specialized branch of artificial intelligence, enables machines to interpret and analyze visual information from images and video. Through algorithms designed to replicate elements of human visual perception, these systems can extract meaningful features, detect patterns, and support decision-making based on visual inputs. Within cardiovascular fitness and rehabilitation, computer vision technologies are increasingly used to track movement quality, assess posture, and deliver real-time corrective feedback, thereby improving the personalization, precision, and effectiveness of exercise interventions.³⁷

Computer vision technologies can evaluate a user's posture and movement patterns during exercise by applying pose-estimation methods that map joint positions and track body alignment. These systems help ensure that movements are performed safely and correctly, lowering injury risk while improving the overall effectiveness of exercise routines. Virtual fitness platforms increasingly use webcams or smartphone cameras to monitor a participant's form and deliver immediate corrective feedback, supporting safer and more effective cardiovascular workouts.³⁷

Computer vision technologies support home-based cardiac rehabilitation by enabling clinicians to remotely evaluate exercise performance with a level of detail comparable to in-clinic supervision. Through video-based analysis of posture and movement, healthcare providers can review recorded sessions, assess adherence to prescribed routines, and modify exercise programs as needed. This approach ensures continuous oversight and individualized guidance, extending high-quality rehabilitative care beyond traditional clinical environments.³⁷

Mobile health applications in exercise, fitness, and cardiovascular rehabilitation

One of the most common functions of mobile health applications is the delivery of health education and coaching. These platforms provide evidence-based content on exercise techniques, chronic disease self-management, nutrition, and general wellness, supporting informed decision-making and sustained engagement in healthy behaviors. In parallel, mobile health tools facilitate symptom monitoring by enabling users to record and communicate changes in health status, supporting timely self-care and clinical intervention.³⁸ Remote patient monitoring meta-analyses further show that wearable biosensors enable continuous health assessments across diverse clinical populations, although impacts on clinical outcomes vary by condition and intervention design.³⁹

By integrating biometric sensor data with user-reported information, mobile health applications can associate symptoms with real-time physiologic measures, allowing emerging patterns to be identified proactively by patients and clinicians. Integration with wearable technologies—including smartwatches, fitness trackers, and advanced biosensors—has further enhanced the precision of exercise monitoring and behavioral feedback. These devices continuously capture metrics such as heart rate, electrocardiographic signals, step counts, and sleep patterns. When synchronized within mobile health ecosystems, these data streams provide a more comprehensive view of individual health trajectories, supporting personalized exercise prescriptions and adaptive rehabilitation plans.⁴⁰ Moreover, recent systematic reviews demonstrate that wearable activity trackers effectively support physical activity improvements not only in adults but also in children and adolescents, underscoring broad applicability across the lifespan.^{39–41}

Mia Health's Activity Quotient (AQ) represents a novel and scientifically grounded approach within this landscape. Unlike traditional physical activity metrics that treat all minutes of activity equally, AQ is based on the fundamental principle that *not all minutes yield the same health benefit*. It is, to our knowledge, the only validated physical activity metric that explicitly weights activity intensity in proportion to its empirically observed association with health outcomes. AQ uses continuous heart rate data from the past seven days to quantify an individual's total "health-effective" physical activity, translating it into a single, personalized score that reflects how much one's exercise contributes to improved health. This concept is supported by a landmark 2025 UK Biobank study,⁴² including approximately 75,000 participants, which demonstrated how exercise intensity strongly influences both disease risk and mortality. The study showed that to achieve the same health benefit as one minute of high-intensity exercise (corresponding to 15–17 on the Borg scale, or ~ 85% of maximal heart rate), individuals would need to perform many more minutes of moderate-intensity and/or low-intensity activity across a range of health outcomes (Fig. 2).

Backed by decades of health data from over 500,000 individuals, AQ

EVERY MINUTE COUNTS BUT SOME COUNT MORE

Outcome	Vigorous activity	Moderate activity	Light activity
T2D incidence	1x	9x	94x
CVD mortality	1x	8x	73x
Major CVD events	1x	5x	86x
All-cause mortality	1x	4x	53x
Cancer mortality	1x	4x	156x

Fig. 2. Estimation of how much moderate- or low-intensity activity is required to achieve the same health benefit as one minute of vigorous exercise (15–17 on the Borg scale). Figure made from results presented in Biswas et al., 2025.⁸²

scores above 25 have been associated with measurable health benefits, while maintaining a weekly AQ of 100 or more is linked to optimal reductions in CVD and all-cause mortality.⁴³ By tailoring exercise guidance to an individual's age, sex, and physiological response, AQ provides a more meaningful and personalized target for health improvement – offering a smarter alternative to traditional “steps” or “minutes” metrics. By incorporating elements such as gamification, progress tracking, and personalized recommendations, AQ fosters long-term behavioral changes, making exercise both measurable and motivating. For healthcare professionals, AQ offers a scalable solution for remote patient monitoring and digital cardiac rehabilitation, ensuring that patients remain active and engaged outside the clinical setting. AQ bridges the gap between exercise science and digital health, offering a highly personalized, accessible solution for cardiovascular care. Whether utilized by individuals aiming to improve their fitness or by healthcare professionals seeking to enhance patient outcomes, AQ exemplifies how technology can be harnessed to promote health and prevent disease in the digital age.

Early clinical studies have shown excellent results for AQ-based exercise guidance, with markedly higher adherence and greater improvements in cardiorespiratory fitness and quality of life compared to standard exercise recommendations. The large-scale NorEx trial is currently testing AQ-guided training in approximately 9500 participants with prior myocardial infarction, evaluating its effectiveness on recurrent cardiovascular events and mortality over a five-year follow-up period.⁴⁴

Despite their wide-ranging benefits, several challenges must be addressed to optimize the effectiveness of mobile health apps. Data privacy and security remain paramount, as these platforms often collect sensitive health information. Robust encryption, compliance with regulatory standards (e.g., the Health Insurance Portability and Accountability Act [HIPAA] and the General Data Protection Regulation [GDPR]), and transparent data policies are essential to protect users. Additionally, sustained user engagement can be difficult to achieve.

The role of virtual reality in exercise training and cardiovascular health

Virtual reality in cardiovascular rehabilitation

VR has emerged as a promising adjunct to cardiovascular rehabilitation, offering immersive environments that enhance both functional outcomes and patient experience. Evidence from a 2025 systematic review and meta-analysis demonstrates that VR-enhanced cardiac rehabilitation significantly improves cardiopulmonary function, including increases in six-minute walk distance (mean difference 34.9 m) and

moderate gains in quality of life, outperforming standard rehabilitation approaches across several functional domains.⁴⁵

Similarly, a 2022 meta-analysis of randomized controlled trials found that VR-based interventions improved exercise capacity by nearly 50 m while also reducing emotional tension, psychological stress, and depressive symptoms among individuals undergoing cardiac rehabilitation.⁴⁶ Beyond physiological benefits, VR meaningfully enhances patient engagement—an important determinant of adherence and long-term rehabilitation success. A 2023 exploratory analysis reported that immersive VR therapy reduced anxiety and depression in cardiovascular patients, with comparable benefits across younger and older adults, suggesting that VR supports emotional involvement and attentiveness throughout therapy sessions.⁴⁷

Consistent with these findings, a systematic review of randomized controlled trials showed that semi-immersive and non-immersive VR systems, including Wii- and Kinect-based platforms, improved functional performance measures such as six-minute walk distance, peak heart rate, and peak oxygen consumption during phases II and III of cardiac rehabilitation.⁴⁸ In addition, a 2026 meta-analysis of telemedicine-based exercise rehabilitation—much of which incorporates VR or VR-adjacent digital platforms—demonstrated significant improvements in exercise capacity, physical activity levels, depressive symptoms, and cardiovascular risk markers compared with usual care.⁴⁹

Collectively, this body of evidence supports VR as a clinically meaningful and engaging modality capable of advancing contemporary cardiovascular rehabilitation through improvements in physiological capacity, psychological well-being, and patient adherence.

VR applications, described in Table 2, illustrate how immersive software tools translate analytic capability into patient-facing interventions. One of the most promising capabilities of VR in cardiovascular care is its potential to promote sustained physical activity through enhanced engagement. By incorporating game-based elements such as point scoring, rewards, and virtual competition, VR can reduce perceived exertion and improve exercise enjoyment. Virtual cycling platforms such as Zwift exemplify this approach by integrating heart rate monitoring with simulated environments and social competition, resulting in increased engagement and longer exercise duration.⁵⁰

Multiple studies indicate that VR-based rehabilitation programs are safe, well tolerated, and comparable to traditional in-person cardiac rehabilitation for improving functional capacity, quality of life, and adherence.^{51–53} Immersive platforms designed around functional fitness tasks, such as those simulating real-world movement challenges, have demonstrated feasibility and acceptability in rehabilitation settings.⁵⁴

Despite this promise, important challenges remain. The cost of VR hardware and software can limit accessibility, particularly in underserved or low-resource settings. In addition, some users experience cybersickness, including dizziness or nausea, which may constrain session duration and adherence.⁵⁵ Clinical integration is also evolving, as standardized protocols for prescribing VR-based interventions are limited and formal training opportunities for clinicians remain sparse.

Looking ahead, advances in AI and sensor technology are expected to further expand the utility of VR in exercise and rehabilitation. AI-enabled VR platforms can dynamically adapt exercise intensity using biometric feedback, supporting safer and more personalized interventions. Emerging multi-user VR environments may facilitate social interaction and group-based rehabilitation, while integration with wearable sensors enables comprehensive cardiovascular monitoring across home and clinical settings.^{56,57}

The role of augmented and mixed reality in exercise training and cardiovascular health

Augmented reality (AR) and mixed reality are emerging immersive technologies that enhance engagement and effectiveness in exercise training by overlaying digital elements onto the physical environment.

These systems deliver real-time instructional cues and personalized coaching by superimposing virtual trainers or guidance onto a user's surroundings. Visual and auditory feedback on movement quality, heart rate, and exercise intensity can be provided during activity, while customizable and often gamified environments improve motivation and adherence. Integration with wearable physiologic sensors further enables continuous biometric monitoring, supporting individualized exercise prescription and progression.⁵⁹

In cardiovascular rehabilitation, augmented and mixed reality-assisted programs offer safe and scalable options for supervised exercise outside traditional clinical settings. These technologies support remote monitoring while preserving key elements of in-person supervision, including feedback on exercise form and intensity. A randomized clinical trial demonstrated that phase I cardiac rehabilitation delivered using AR significantly improved patient self-efficacy following coronary artery bypass graft surgery, highlighting the potential of AR-based interventions in early rehabilitation phases.⁵⁸

The integration of AR and mixed reality into cardiovascular care offers several advantages. Gamified environments promote sustained motivation and long-term adherence, while remote delivery expands access for individuals in rural or underserved settings. Personalization is achieved through real-time physiologic feedback, allowing exercise programs to be dynamically adjusted to ensure safety and effectiveness. In addition, immersive educational experiences embedded within AR and mixed platforms may facilitate behavior change by deepening patient engagement and understanding of cardiovascular health.⁵⁹

Looking ahead, continued innovation is expected to further strengthen the role of AR and mixed reality in cardiovascular prevention and rehabilitation. Advances in wearable integration will support more comprehensive physiologic monitoring, while artificial intelligence-enabled analytics will allow dynamic adjustment of exercise prescriptions in response to real-time data. The development of clinical-grade, and potentially regulatory-approved, AR and mixed reality applications for rehabilitation and chronic disease management may help bridge the gap between digital health tools and conventional care models, enabling more immersive, patient-centered cardiovascular care ecosystems.⁵⁹

Chatbots in cardiovascular fitness and rehabilitation

Chatbots are AI-driven conversational agents that deliver automated, personalized support via text or voice. In cardiovascular fitness and rehabilitation, chatbots are increasingly used to support exercise engagement, education, and self-management beyond traditional clinical encounters.

Chatbots can provide individualized exercise reminders, goal tracking, and behavioral feedback based on user-reported data and wearable inputs. Evidence from randomized trials and systematic reviews suggests that chatbot-supported interventions can improve adherence to home-based exercise and rehabilitation programs, particularly when integrated within broader digital health ecosystems.^{60,61} In parallel, chatbots offer accessible education on exercise safety, heart rate targets, and lifestyle modification, supporting health literacy and self-efficacy.⁶² Gamification strategies, including goal setting and positive reinforcement, are commonly embedded to promote sustained behavior change.⁶²

When integrated with mobile health platforms and wearable devices, chatbots can facilitate symptom reporting and passive data capture, enabling clinicians to review longitudinal trends and adjust exercise prescriptions as needed. In most implementations, data are reviewed asynchronously rather than in real time, consistent with established remote patient monitoring workflows and existing clinical practice patterns.¹⁴

Despite their promise, chatbots have important limitations in cardiovascular care. Current systems may lack clinical context awareness, perform inconsistently in complex or ambiguous scenarios, and risk

generating incomplete or misleading guidance if not appropriately constrained.^{15,30} Data privacy, transparency, and algorithmic bias remain critical concerns, particularly when conversational agents are trained on large, non-curated datasets.^{15,36} Engagement also varies across populations, with lower uptake observed among individuals with limited digital literacy or access to technology.³⁰

Effective implementation requires positioning chatbots as adjuncts—not replacements—for clinician-led care. Best practices include clearly defining chatbot scope, incorporating escalation pathways to human oversight, using evidence-based content, and ensuring transparency around data use and system limitations. Hybrid models that combine chatbot support with clinician or coach involvement appear most effective for balancing scalability, safety, and patient trust.^{30,62}

The role of social networks in exercise, fitness, and rehabilitation

Social networks—both interpersonal and digital—are important determinants of engagement, adherence, and sustained behavior change in exercise, fitness, and cardiovascular rehabilitation. Behavioral and social science research consistently shows that social support, accountability, and normative influence strongly shape long-term adherence to lifestyle interventions, including physical activity.⁶³ These mechanisms are particularly relevant in cardiovascular prevention and rehabilitation, where behavior change must be maintained over time.

Digital health platforms increasingly incorporate structured social networking features within mobile applications, wearable ecosystems, and telehealth programs. Quantitative evidence indicates that social connectivity meaningfully modifies outcomes. Large-scale observational analyses of fitness tracking platforms demonstrate that individuals embedded in active social networks accumulate higher levels of physical activity and show greater persistence compared with socially isolated users, consistent with “exercise contagion” effects across networks.^{64,65} Randomized trials in weight management and lifestyle interventions further show that group-based or socially supported programs achieve greater adherence and weight loss than app-only or self-directed approaches.⁶⁶

Within rehabilitation and chronic disease management, digital social networks also function as extensions of formal education and coaching. Moderated peer communities facilitate the exchange of practical strategies, normalize challenges related to exercise and recovery, and reinforce motivation. Systematic reviews indicate that digital interventions incorporating peer interaction are associated with increased physical activity, improved adherence, and higher self-efficacy.^{67,68} In cardiac rehabilitation, programs integrating peer support—through group-based digital platforms or hybrid home-based models, demonstrate higher participation and completion rates than conventional approaches alone.⁶⁹

Beyond purpose-built health platforms, social media-based communities have emerged as influential venues for cardiovascular health discourse. Analyses of platforms such as Twitter and Reddit show that patients frequently seek experiential knowledge, emotional support, and practical guidance related to cardiovascular disease and rehabilitation.⁷⁰ Work by Somani and colleagues highlights both the reach and limitations of these forums, underscoring their potential to complement clinical care while also raising concerns regarding misinformation and variable content quality.⁷¹

Despite these benefits, challenges remain. Privacy concerns, data security risks, and the potential spread of misinformation may limit participation or undermine evidence-based care. Effective implementation therefore requires active moderation, clear governance, and integration with trusted clinical guidance to maximize benefit while minimizing harm.

The role of gaming in exercise, fitness, and rehabilitation

Exergames, or exercise-based video games, combine physical activity with interactive digital experiences and have emerged as engaging tools for promoting cardiovascular fitness. Titles such as *Just Dance*, *Ring Fit Adventure*, and virtual reality-based platforms like *Beat Saber* require dynamic, whole-body movements that elevate heart rate and can achieve moderate-to-vigorous intensity physical activity. Evidence from systematic reviews and controlled studies indicates that exergaming is associated with improvements in cardiorespiratory fitness and modest reductions in blood pressure, supporting its potential role in cardiovascular health promotion.^{55,72}

Gamification—the application of game-design elements such as points, challenges, and leaderboards in non-game contexts—has been increasingly integrated into fitness and rehabilitation programs to enhance engagement and adherence. Digital fitness platforms including Strava and Nike Training Club incorporate these features to reinforce motivation, goal attainment, and social accountability. Observational and interventional studies suggest that gamified approaches are associated with higher physical activity levels and greater persistence compared with non-gamified interventions.²⁷

In cardiac rehabilitation, gamified interventions have been developed to improve participation and adherence to lifestyle modification programs. For example, the *MyHeartMate* application employs goal setting, rewards, and narrative elements to support behavior change among individuals with cardiovascular disease and has demonstrated improved engagement and adherence compared with usual care.²⁷

Serious games—designed explicitly for health or rehabilitation purposes rather than entertainment—have similarly been used to transform prescribed exercise into interactive tasks. Studies suggest that serious games can improve motivation, increase participation in rehabilitation programs, and support functional gains, although evidence for long-term clinical outcomes remains limited and continues to evolve.⁷³

The integration of wearable sensors and artificial intelligence into gaming platforms further enhances personalization and safety. Wearable devices enable continuous monitoring of physiologic parameters such as heart rate and movement patterns, while AI-driven algorithms can dynamically adjust exercise intensity to maintain users within target training zones. These adaptive features have been shown to support individualized exercise delivery, reduce inappropriate exertion, and improve the feasibility of home-based cardiovascular training across diverse user populations.^{74,75}

Implementation considerations for patients and providers

Effective implementation of digital exercise and rehabilitation technologies requires alignment across patients, clinicians, and health systems. For patients, usability, digital literacy, cultural relevance, and perceived value are central determinants of adoption and sustained engagement. Interventions that incorporate structured onboarding, simplified user interfaces, and hybrid models combining digital tools with human coaching or clinical oversight demonstrate higher uptake and adherence, particularly among older adults and individuals with chronic disease.^{74,75}

For providers, successful implementation depends on seamless workflow integration, clarity regarding clinical responsibility, and confidence in the validity and actionability of incoming data. Implementation science frameworks, including the RE-AIM model, emphasize the importance of reach, effectiveness, adoption, implementation fidelity, and maintenance when deploying digital health interventions in real-world clinical settings.⁷⁶ Embedding digital tools within existing care pathways—rather than positioning them as standalone adjuncts—appears critical for sustainability, clinician acceptance, and long-term impact.

Cost-effectiveness, reimbursement, and policy considerations

A growing body of evidence suggests that digital and home-based exercise and rehabilitation programs can be cost-effective alternatives to traditional center-based models, particularly in cardiac rehabilitation and secondary prevention. Randomized trials and economic evaluations demonstrate that home-based and technology-supported cardiac rehabilitation achieve comparable improvements in functional capacity and quality of life at similar or lower overall costs, largely by reducing travel burden, facility utilization, and staffing requirements.^{55,77,78} For example, Kraal et al. reported that home-based cardiac rehabilitation was clinically non-inferior to center-based programs and associated with favorable cost profiles, particularly when adherence was maintained.⁶⁹ Similarly, Maddison et al. demonstrated that real-time telerehabilitation achieved improvements in cardiovascular risk factors with acceptable incremental cost-effectiveness ratios relative to usual care.⁷⁷

Policy statements from professional societies have increasingly endorsed remote and hybrid models of cardiac rehabilitation, particularly following the COVID-19 pandemic. The American Heart Association and American College of Cardiology have highlighted home-based cardiac rehabilitation as a viable alternative to traditional programs when appropriately structured and monitored.⁷⁹ In parallel, the European Society of Cardiology Working Group on e-Cardiology has emphasized the economic and scalability advantages of digitally supported models of care, while noting the need for robust health economic evaluations across diverse health systems.⁷⁸

Despite promising data, important gaps remain. Many cost-effectiveness studies are limited by short follow-up durations, single-country analyses, or narrowly defined payer perspectives. Few studies incorporate long-term cardiovascular event reduction, workforce productivity, or societal cost considerations into their models. Additionally, reimbursement pathways for remote patient monitoring, digital therapeutics, and virtual rehabilitation remain heterogeneous across jurisdictions.^{79,80} While some payers have introduced billing codes for remote monitoring and virtual supervision, coverage often lags behind technological capability, creating uncertainty for health systems seeking sustainable implementation.

Future research should prioritize long-term economic modeling, comparative effectiveness across hybrid and fully digital models, and evaluation of cost-effectiveness in underserved populations where travel and access barriers are greatest. Policy frameworks aligned with value-based care principles—rewarding outcomes, prevention, and continuity rather than volume—will likely determine the scalability and sustainability of digital exercise-based cardiovascular interventions.

Data scale, signal, and clinical usability

Digital exercise and monitoring technologies generate large volumes of continuous physiologic and behavioral data, creating both opportunity and burden for clinicians and health systems. While high-frequency data streams enable personalization and longitudinal assessment, poorly filtered or excessive data can overwhelm providers and reduce clinical utility.⁸¹ Distinguishing meaningful signal from noise is therefore central to effective deployment.

Strategies to improve data usability include aggregation into clinically interpretable metrics, trend-based visualization rather than point estimates, threshold-triggered alerts, and selective escalation of clinically relevant changes. Artificial intelligence, enabled analytics and rule-based systems can further support prioritization by identifying patterns associated with risk, adherence, or deterioration, reducing the need for continuous manual review.²¹ Critically, digital systems should deliver actionable insights rather than raw data, as frequent notifications without clear clinical relevance may discourage provider engagement and undermine adoption.

Across these domains, digital cardiovascular care emerges not as a collection of isolated technologies but as an integrated ecosystem in

which analytic systems, connected devices, immersive platforms, and implementation frameworks function interdependently. The effectiveness of any individual tool depends on its alignment with infrastructure, clinical workflows, and patient context.

Summary

Digital health technologies have matured from adjunct tools into integral components of contemporary exercise-based cardiovascular prevention and rehabilitation. Evidence across mobile health platforms, wearable sensors, remote monitoring systems, and AI-enabled analytics demonstrates their capacity to improve physical activity engagement, personalize exercise prescription, and extend care beyond traditional clinical settings. When thoughtfully implemented, digital interventions can achieve outcomes comparable to center-based models, support long-term adherence, and enhance access for populations historically underserved by conventional care. However, their effectiveness depends on purposeful integration into clinical workflows, attention to equity and digital literacy, reimbursement alignment, and the transformation of high-volume data into actionable insights. Collectively, the evidence supports a shift toward hybrid, technology-enabled models of cardiovascular care in which digital tools augment, rather than replace, human clinical judgment, placing effective, scalable exercise support directly into patients' everyday lives.

CRedit authorship contribution statement

Patrick Dunn: Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Rachael Charbonneau:** Writing – original draft, Visualization, Conceptualization. **Cemal Ozemek:** Writing – review & editing, Writing – original draft. **Ross Arena:** Writing – review & editing, Writing – original draft, Formal analysis. **Ulrik Wisløff:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

Professor Wisløff is the inventor of AQ. An NTNU-spin-off-company (Mia Health inc) holds the IP rights for AQ that develops applications that utilize data from diverse heart rate monitors to display AQ for users. Due to the potential conflict of interest, Wisløff do not take part in any of the statistical analysis in any AQ-studies. All other authors declare no competing interests.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

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